

Problem set 1 – Econ 5030

1. Defs: $x \succ y \iff x \succsim y \wedge \neg(y \succsim x)$; $x \sim y \iff x \succsim y \wedge y \succsim x$.
 - (a) $x \succ x$ needs $x \succsim x$ and $\neg(x \succsim x)$. Contradiction.
 - (b) $x \succsim y, y \succsim z \Rightarrow x \succsim z$ (trans.). Suppose $z \succsim x$: then $z \succsim x, x \succsim y \Rightarrow z \succsim y$, contradicts $y \succ z$. So $\neg(z \succsim x)$, hence $x \succ z$.
 - (c) Completeness $\Rightarrow x \succsim x$. So $x \succsim x \wedge x \succsim x \Rightarrow x \sim x$.
 - (d) $x \succsim y, y \succsim z \Rightarrow x \succsim z$; $z \succsim y, y \succsim x \Rightarrow z \succsim x$. So $x \sim z$.
 - (e) $x \sim y \iff x \succsim y \wedge y \succsim x \iff y \succsim x \wedge x \succsim y \iff y \sim x$.
 - (f) $x \succsim y, y \succsim z \Rightarrow x \succsim z$. Suppose $z \succsim x$: then $y \succsim z, z \succsim x \Rightarrow y \succsim x$, contradicts $x \succ y$. So $x \succ z$.
2. (a) Completeness: pick $x_2 \in \{a, b\}$ with $x_2 \succsim a, x_2 \succsim b$. Compare x_2 vs c ; pick the better one x_3 . If $x_3 = c$ then $c \succsim x_2 \succsim a, b \Rightarrow c \succsim a, b$ (trans.). Either way $x_3 \succsim a, b, c$. Repeat with d : get $x_4 \succsim a, b, c, d$. So $x_4 \in C(X, \succsim) \neq \emptyset$.
 - (b) $x, y \in C(X, \succsim) \Rightarrow x \succsim y$ (since $x \succsim$ all) and $y \succsim x$ (since $y \succsim$ all) $\Rightarrow x \sim y$.
3. (a) Let $L(x) = \{z \in X : x \succsim z\}$, $u(x) = |L(x)|$. Claim $x \succsim y \iff u(x) \geq u(y)$.
 - ($\Rightarrow$) $x \succsim y, z \in L(y) \Rightarrow y \succsim z \Rightarrow x \succsim z$ (trans.) $\Rightarrow z \in L(x)$. So $L(y) \subseteq L(x)$, $u(y) \leq u(x)$.
 - ($\Leftarrow$) Suppose $u(x) \geq u(y)$ but $\neg(x \succsim y)$. Completeness $\Rightarrow y \succsim x$, so $L(x) \subseteq L(y)$ by above. But $y \in L(y)$ ($y \succsim y$) and $y \notin L(x)$. So $L(x) \subsetneq L(y) \Rightarrow u(x) < u(y)$. Contradiction.
- (b) $[0, 1]^2$ is uncountable; complete + transitive alone may have too many strict levels to embed in $\mathbb{R}$. Need *continuity* (contour sets closed) $\Rightarrow$ Debreu.

Example: lexicographic $x \succsim y \iff x_1 > y_1$ or $(x_1 = y_1, x_2 \geq y_2)$. Complete, transitive. Suppose u represents it. $\forall t \in [0, 1]: (t, 1) \succ (t, 0) \Rightarrow I_t = (u(t, 0), u(t, 1)) \neq \emptyset$. $t < t' \Rightarrow (t', 0) \succ (t, 1) \Rightarrow I_t \cap I_{t'} = \emptyset$. Uncountably many disjoint open intervals, each contains a rational $\Rightarrow$ injection $[0, 1] \rightarrow \mathbb{Q}$. Contradiction.
4. $u = x_1^{1/2} x_2^{1/2}$.
 - (a) Interior: $x_i = 0 \Rightarrow u = 0$, but $(\frac{w}{2p_1}, \frac{w}{2p_2})$ affordable with $u > 0$. Budget binds: u strictly increasing, so if $p \cdot x < w$ raise x_1 by ε , still affordable, u up.

FOC: $\mathcal{L} = x_1^{1/2} x_2^{1/2} - \lambda(p_1 x_1 + p_2 x_2 - w)$

$$\frac{1}{2} x_1^{-1/2} x_2^{1/2} = \lambda p_1, \quad \frac{1}{2} x_1^{1/2} x_2^{-1/2} = \lambda p_2 \Rightarrow \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow p_1 x_1 = p_2 x_2.$$

Budget: $2p_1 x_1 = w$. So

$$D(p, w) = \left(\frac{w}{2p_1}, \frac{w}{2p_2} \right), \quad v(p, w) = \frac{w}{2\sqrt{p_1 p_2}}.$$

(Sufficient: $\ln u$ strictly concave, budget set convex.)

(b) $x_1 = \frac{80}{8} = 10$, $x_2 = \frac{80}{2} = 40$, $v = \frac{80}{2\sqrt{4}} = 20$. Check $\sqrt{10 \cdot 40} = 20$.

(c) Walras: $4(10) + 1(40) = 80 = w$. ✓

$\lambda = 2$: $D(8, 2, 160) = (\frac{160}{16}, \frac{160}{4}) = (10, 40) = D(4, 1, 80)$. ✓

5. $u = \min\{x_1, 2x_2\}$.

(a) Any affordable x , $t = \min\{x_1, 2x_2\}$: $x_1 \geq t$, $x_2 \geq t/2$, so

$$w \geq p_1x_1 + p_2x_2 \geq p_1t + p_2\frac{t}{2} \Rightarrow t \leq \frac{2w}{2p_1 + p_2}.$$

Bound attained (uniquely) iff $x_1 = 2x_2$ and $p \cdot x = w$:

$$D(p, w) = \left(\frac{2w}{2p_1 + p_2}, \frac{w}{2p_1 + p_2} \right), \quad v(p, w) = \frac{2w}{2p_1 + p_2}.$$

Econ: goods used in fixed ratio $x_1 = 2x_2$; extra units of either are wasted. Buy “kits” $(2, 1)$ at price $2p_1 + p_2$; # kits = $w/(2p_1 + p_2)$. No substitution, relative prices don’t change proportions.

(b) $2p_1 + p_2 = 8$: $x_1 = \frac{160}{8} = 20$, $x_2 = \frac{80}{8} = 10$, $v = 20$. Check: $3(20) + 2(10) = 80$, $\min\{20, 20\} = 20$.

(c) Convex: $\{u \geq t\} = \{x_1 \geq t\} \cap \{x_2 \geq t/2\}$, intersection of half-planes $\Rightarrow$ convex. (Or: min of linear $\Rightarrow$ concave $\Rightarrow$ quasiconcave.)

Not strictly: $x = (2, 1)$, $y = (4, 1)$, $u(x) = u(y) = 2$, $x \sim y$, $x \neq y$. $z = \frac{1}{2}x + \frac{1}{2}y = (3, 1)$, $u(z) = 2$, so $z \sim x$, not $z \succ x$.

Singleton: strict convexity is sufficient, not necessary. Flat parts of the L have $x_1 \neq 2x_2$: wasted spending, dominated by (a). Only the kink $x_1 = 2x_2$ can be optimal, and it meets the budget line once. Singleton $\Rightarrow$ convex-valued.